

An Approach to a Case of Cough: Lung Abscess, Bronchiectasis & Cystic Fibrosis

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Lung_abscess_Cystic_Fibrosis__Bronchiactasis_updated.pdf (44 slides) · Refs: Step-Up 5th ed.; Harrison 20th ed.; AMBOSS; Cecil 10th ed.; First Aid IM Boards 4th ed.

Legend: ★ = high-yield · [added] = not on slides. Images marked "Slide N" come from the lecture; "Added diagram" = drawn for these notes.

🔗 **Bridge from the COPD and Asthma lectures:** those were **obstructive** airway diseases. This lecture covers **suppurative** (pus-forming) disease, and the links are direct: **COPD, α 1-antitrypsin deficiency, aspiration and Kartagener syndrome** (all named in the COPD lecture) are **causes of bronchiectasis**, and **CF** is the classic genetic cause. The CF defect is a **faulty ATP-gated Cl^- channel**, the kind of membrane transport covered in Physiology Lectures 02-03.

Suppurative lung disease: three diseases side by side

	LUNG ABSCESS	BRONCHIECTASIS	CYSTIC FIBROSIS
What it is	Localised pus + necrosis in lung parenchyma	Permanent dilatation and destruction of bronchial walls	AR CFTR defect → thick secretions in all exocrine glands
Key cause	Aspiration (oral anaerobes) septic emboli, IVDU (S. aureus)	Infection/inflammation + poor clearance (40-50% idiopathic)	Δ F508 (Phe508 deletion) misfolded Cl^- channel
Sputum	FOUL-smelling purulent (↑ lying on healthy side)	LARGE volume mucopurulent daily	Chronic productive cough (Pseudomonas, Aspergillus)
Imaging	Thick-walled cavity with AIR-FLUID LEVEL (CT confirms)	HRCT: TRAM-TRACK + SIGNET-RING sign	Central cystic bronchiectasis, mucus plugs, hyperinflation
Key test	Sputum/BAL Gram stain + culture (blood cultures often -)	HRCT; then find the cause (Igs, Aspergillus IgE, AAT)	Sweat $\text{Cl}^- \geq 60 \text{ mmol/L}$ + CFTR genotyping
Clubbing	Chronic abscess	Uncommon	Yes (chronic hypoxia)
Treatment	Anaerobic cover: ampicillin-sulbactam / carbapenem / clindamycin, 3-6 weeks	Physio, mucolytics, vaccines; antibiotics $\geq 3 \text{ mo}$ if ≥ 3 exacerbations/yr	Airway clearance, DNase, hypertonic saline, anti-Pseudomonas Abx, enzymes
If it fails	Drain (bronchoscopic/percut.)	Lobectomy (bleeding)	Double-lung transplant

Links: CF → bronchiectasis · bronchiectasis → abscess risk · Kartagener (ciliary dyskinesia) → bronchiectasis + situs inversus + sinusitis

Added diagram: lung abscess vs bronchiectasis vs cystic fibrosis

1. Title / Objectives

Domain	Objective	Likely tested as
Knowledge	★ Define bronchiectasis, lung abscess, CF	Definitions
Knowledge	★ Pathophysiology: the vicious cycle of bronchiectasis, abscess formation, CFTR dysfunction	Cole's cycle; sweat vs airway Cl^-
Knowledge	Congenital, infectious, obstructive and systemic causes	Cause lists
Knowledge	★ Clinical features and multi-organ complications: respiratory failure, clubbing, pancreatic insufficiency, CF liver disease, infertility	CF organ systems

Domain	Objective	Likely tested as
Knowledge	★ Management: airway clearance, CFTR modulators, anti-Pseudomonas antibiotics, replacement therapy	Drug choices
Skills	★ Imaging signs on CXR/HRCT; differential of chronic cough, dyspnoea, haemoptysis; choose tests; plan acute vs chronic care	Air-fluid level; tram-track; signet ring; sweat test
Values	Burden of chronic illness; nutrition; immunisation	

Flow: Suppurative lung disease → **Lung abscess** (definition, organisms, risks, pathophysiology, features, diagnosis, location, treatment) → **Bronchiectasis** (definition, causes, vicious cycle, features, HRCT signs, cause work-up, management) → **Cystic fibrosis** (definition, genetics, sweat vs airway pathophysiology, organ features, diagnosis, management) → Summaries → Cases.

2. Main Content (Lecture Order)

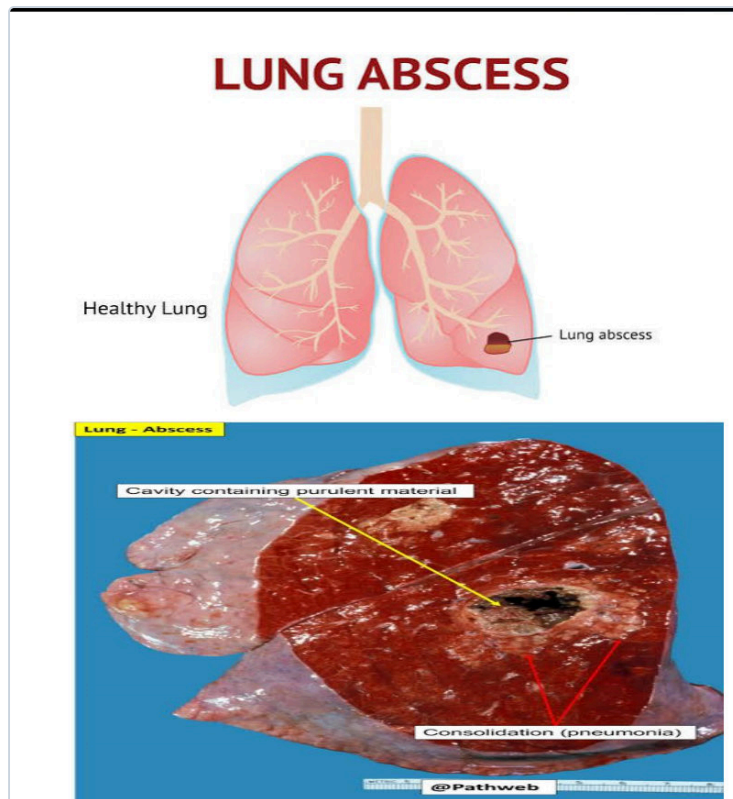
2.1 Suppurative Lung Disease (Slide 5)

- **Suppurative** = pus formation
- ★ **Suppurative lung disease** = chronic inflammation with **formation or discharge of pus** within the **airways or lung tissue**
- **Main types:** 1. **Lung abscess** · 2. **Bronchiectasis** · 3. **Cystic fibrosis** · 4. **Pneumonia**

2.2 Lung Abscess: Definition & Organisms (Slides 6-7)

★ **Definition:** a **localised collection of pus and necrotic tissue** within lung parenchyma caused by **microbial infection**.

Type	Meaning
★ Primary	In normal lung , typically from ★ aspiration
★ Secondary	In immunocompromise or pre-existing disease (e.g., lung neoplasm), or from haematogenous spread (septic emboli, IV drug use)



Slide 6: Lung abscess: diagram and gross specimen

★★ Organisms:

- ★ **Most common: oral anaerobes: Peptostreptococcus, Prevotella, Bacteroides, Fusobacterium**
- Less common aerobes: **microaerophilic and viridans streptococci; S. aureus, Klebsiella pneumoniae, Pseudomonas, group A Strep, S. anginosus, H. influenzae**
- **Parasitic: Entamoeba histolytica**
- **Fungal: Aspergillus, Cryptococcus, Coccidioides, Histoplasma**

Summary slide adds: **Klebsiella in diabetics; TB, M. avium, M. kansasii; Paragonimus westermani.**

2.3 Risk Factors & Pathophysiology (Slides 8-9)

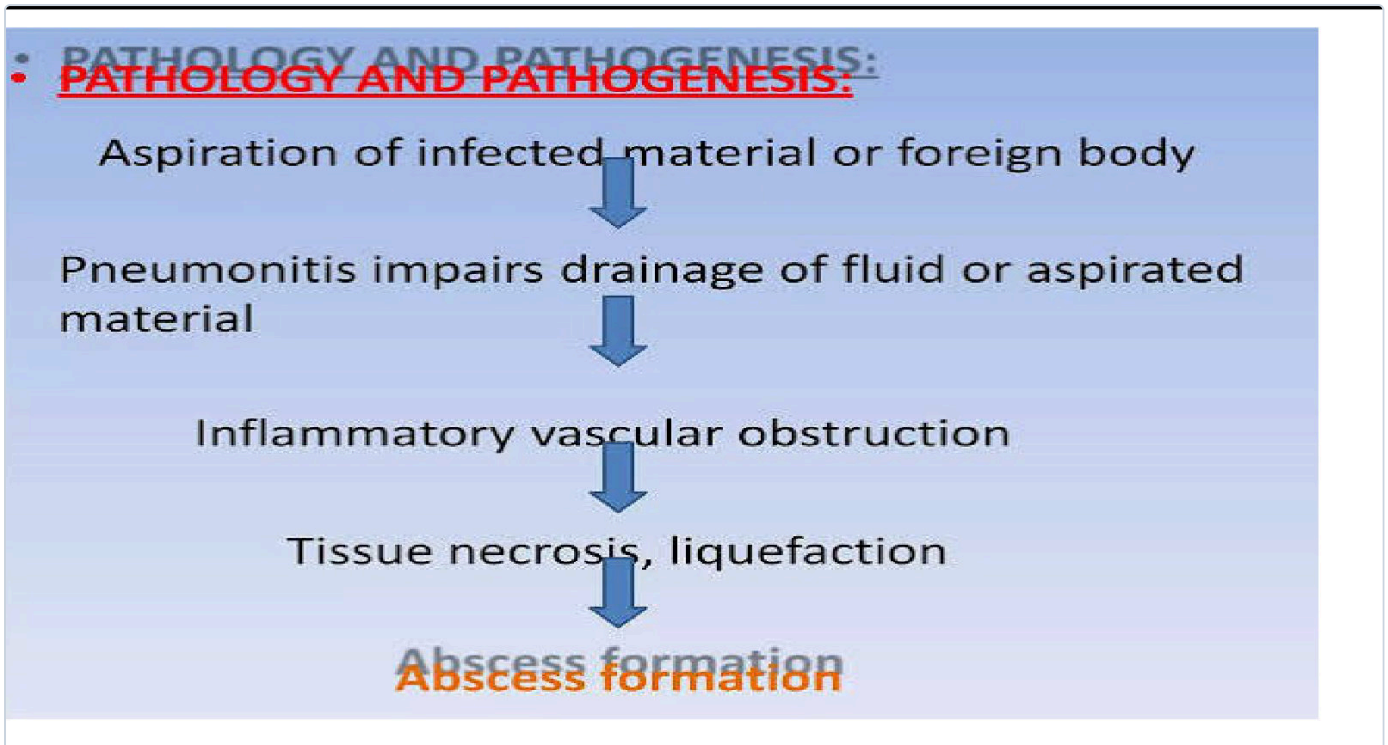
★ Risk factors:

- ★ **Aspiration risk: impaired consciousness** (alcohol [summary slide], seizures [added]), **impaired swallowing** (neurological disease, **vocal-cord paralysis**, oesophageal dysmotility)
- ★ **More oral bacteria: periodontal disease, dental abscess, tonsillitis** (poor dentition)
- ★ **Bronchial obstruction: lung cancer, foreign body, bronchial stenosis**
- **Immunocompromise**
- **Pneumonia, bronchiectasis**
- **Impaired mucus clearance** (e.g., **CF**)
- **Summary slide: IV drug use, large-bore feeding tubes, DM**

★ Pathophysiology sequence (Slide 9):

1. **Aspiration** of infected material or a foreign body
2. **Pneumonitis** impairs drainage of fluid / aspirated material
3. **Inflammatory vascular obstruction**
4. **Tissue necrosis, liquefaction**

5. ★ **Abscess formation** (summary slide: necrosis about 7-14 days after aspiration)

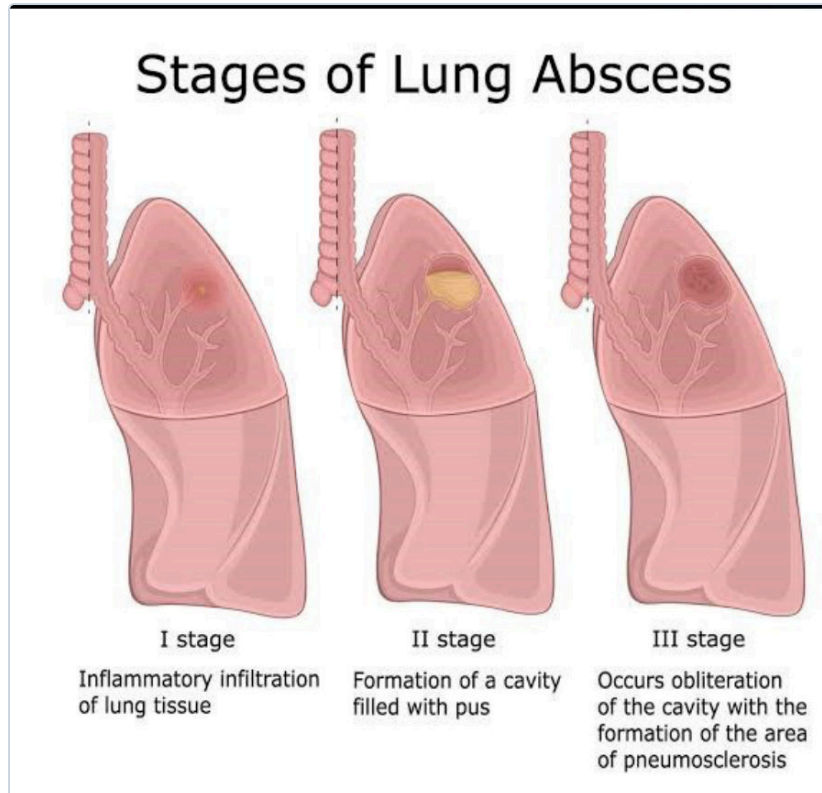


Slide 9: Aspiration → pneumonitis → vascular obstruction → necrosis → abscess

2.4 Clinical Features (Slide 10)

Onset	Duration
Acute	Symptoms < 4-6 weeks
Chronic	Symptoms > 6 weeks

Symptoms: ★ **fever** (~80% > 38 °C) · ★ **cough with foul-smelling purulent sputum** (★ ↑ **when lying on the healthy side**, as the cavity drains [*mechanism added*]) · **night sweats** · **haemoptysis** · anorexia, **weight loss**, fatigue · **pleuritic chest pain** **Signs:** ★ **clubbing in chronic abscess** · **dullness** over the area · **bronchial breath sounds** · poor dentition / gingival disease



Slide 10: Stages: inflammatory infiltration → cavity with pus → cavity obliteration and pneumosclerosis

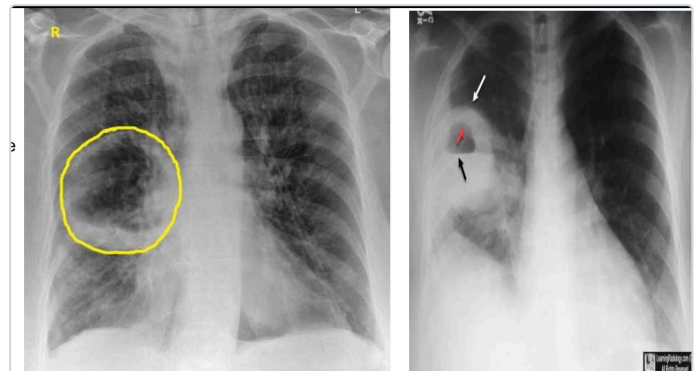
2.5 Diagnosis & Location (Slides 11-12)

Test	Finding / rule
CBC	↑ WBC (± anaemia)
CRP, procalcitonin	↑
Blood culture	★ Often negative in anaerobic infection
★ Sputum / BAL Gram stain and culture	★ Don't wait for results: take samples, start empirical antibiotics immediately , adjust later
HIV screen	If atypical pathogens grow
Bronchoscopy	Suspected malignancy or foreign body
Echocardiography	Suspected endocarditis (septic emboli)
CXR	First imaging (may be non-diagnostic early)
★ CT chest with IV contrast	★ Confirms the diagnosis

★★ **Imaging hallmark: spherical intraparenchymal cavity with thick irregular walls + ★ air-fluid level.**



Slide 11: CXR showing a cavity with an air-fluid level



Slide 12: Abscess location on CXR

★ Location (Slide 12):

- **Aspiration** → typically **unilateral, right-sided** (right main bronchus is wider and more vertical [added])
 - **Right middle lobe**: per slide, aspiration while **prone**
 - **Posterior segments of upper lobes / superior segments of lower lobes**: aspiration while **recumbent (supine)**
- **Haematogenous** → typically ★ **bilateral and multiple**

2.6 Treatment (Slide 13)

★★ Antibiotics:

- ★ **Broad-spectrum with anaerobic cover: ampicillin-sulbactam, carbapenems, or clindamycin**
- **IV drug use** → make sure **gram-positive cocci (S. aureus)** are covered
- Adjust to **culture and response**
- ★ Continue **3-6 weeks** or until clinical and radiological improvement
- *Summary slide*: IV ampicillin-sulbactam; piperacillin-tazobactam (*Pseudomonas*); carbapenems; **IV clindamycin if penicillin-allergic**; switch to oral once stable

★ Interventional therapy:

Indications	Options
Large abscess (<i>summary: > 6 cm</i>)	★ First line: bronchoscopic or image-guided percutaneous drainage
Significant haemoptysis	Rarely: surgical resection (segmentectomy, lobectomy, pneumonectomy)
Poor response : persistent fever/purulent sputum (<i>summary: fever > 10 days</i>), poor radiological resolution	

Summary slide: 10-15% need lobectomy/pneumonectomy if antibiotics fail; a delayed response should prompt a search for **foreign body, cancer or bronchial stenosis**. **Differential:** vasculitis (granulomatosis with polyangiitis), malignancy, aspirated foreign body, bronchostenosis.

LUNG ABSCESS @rav7ks

Etiology:

- necrosis of lung parenchyma by a polymicrobial infection
- most common cause: aspiration pneumonia
- DM risk factor for Klebsiella pneumonia
- Chronic alcoholism
- periodontal disease/ poor dentition/ gingival disease
- IV substance use
- Esophageal dysmotility
- Depressed consciousness
- Large volume feeding tubes

Abscess

Exam:

- poor dentition
- gingival crevice disease
- diminished breath sounds
- rales

Clinical presentations:

- 80% have Fever 38 degrees
- chills, night sweats, fatigue, unexplained weight loss
- Late disease: pleuritic chest pain, hemoptysis, dyspnea, productive cough
- putrid sputum/sour-tasting sputum

Pathophysiology:

Diagnosis:

- CBC leukocytosis, anemia
- Sputum/ Blood cultures
- CRP (non-diagnostic in early disease)
- CXR upright or lat decubitus - air fluid level seen
- Chest CT

Pathogens:

Aerobic Bacteria:

- Staph. Aureus.
- Legionella.
- H. Influenza.
- E. Coli.
- P. Aeruginosa.
- Strep. Pyogenes.
- K. Pneumoniae.

Anaerobic bacteria:

- Bacteroides.
- Fusobacterium.
- Peptostreptococcus.
- Prevotella.

Fungal:

- Aspergillus.
- Blastomyces.
- dematioides.
- Cryptococcus.
- Coccidioides.

Parasite:

- Echinococci Histolytica.
- Paragonimus westermani.

Other: TB

- M. avium.
- M. Kersii.

Differential diagnosis:

- vasculitis (eg. granulomatosis with polyangiitis).
- malignancy.
- aspirated foreign body.
- bronchostenosis.

Management:

Antibiotics

- IV ampicillin-sulbactam
- Piperacillin/Tazobactam: pseudomonas
- carbapenems
- IV clindamycin in case of PCN allergy
- Switch to Oral antibiotics once stable

- 10-15% require lobectomy or pneumonectomy if fail antibiotics

Consider surgical intervention:

- Fever > 10 days
- hemoptysis
- cavitation > 6 cm
- neoplasm, or hemorrhage

Treatment duration:

3 weeks or Continue antibiotic treatment until the chest radiograph shows a small, stable residual lesion or is clear. This generally requires several months of treatment

Delayed response: evaluate for foreign body, cancer, or bronchial stenosis.

Slide 37: Lung abscess summary card

Recap: Lung abscess. Pus + necrosis, mostly from aspiration of oral anaerobes (poor teeth, impaired consciousness/swallowing). Fever, foul sputum, night sweats, weight loss, clubbing if chronic. Cavity with air-fluid level; CT confirms. Start anaerobic-covering antibiotics immediately (ampicillin-sulbactam, carbapenem, clindamycin) for 3-6 weeks. Drain if large, bleeding, or not responding.

Link: An abscess destroys a localised area of parenchyma. Repeated infection and inflammation in the **airways** instead cause permanent bronchial widening: bronchiectasis.

Up next: Bronchiectasis.

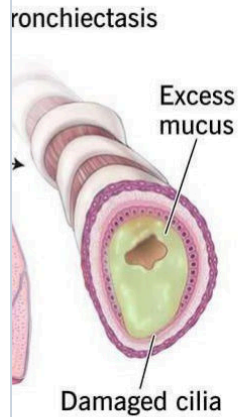
2.7 Bronchiectasis: Definition & Mechanism (Slides 14-15)

★★ **Definition:** permanent, abnormal dilatation and destruction of bronchial walls with chronic infection, sputum production, airway collapse, and ciliary loss/dysfunction → impaired clearance of secretions.

★ **It needs two processes together:**

1. **Local infection or inflammation**, plus
2. **Inadequate clearance of secretions, airway obstruction, or impaired host defence** → permanent airway dilatation

n of bronchial walls with chronic
ry loss/dysfunction leading to



Slide 14: Normal bronchus vs bronchiectasis (excess mucus, damaged cilia)

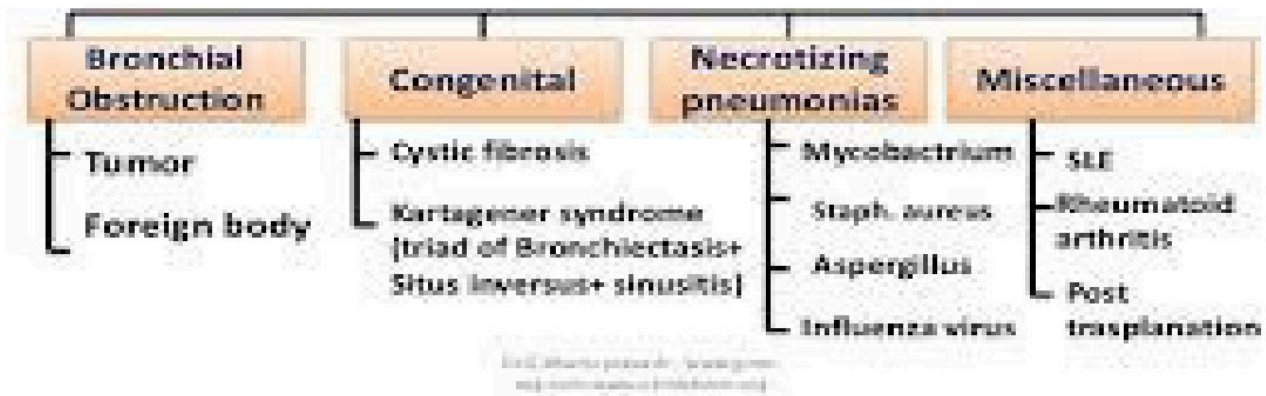
2.8 Causes (Slides 16-17)

Group	Causes
Infection / inflammation	★ Recurrent infection ; necrotising pneumonia (TB/mycobacteria, S. aureus, Aspergillus, influenza); autoimmune disease
★ Clearance defects / mucus plugging	★ Cystic fibrosis · ★ primary ciliary dyskinesia · ★ ABPA · ★ Kartagener syndrome (bronchiectasis + situs inversus + sinusitis) · smoking (poor ciliary motility)
★ Obstruction	COPD, aspiration, tumours, foreign body, α1-antitrypsin deficiency, tracheomalacia, bronchogenic cyst
Immunodeficiency	Humoral (hypogammaglobulinaemia), HIV
Chronic inflammatory disease	Rheumatoid arthritis, Sjögren, Crohn, SLE , post-transplant
★ Idiopathic	★ 40-50% of cases

4. Bronchiectasis

- Abnormal permanent airways dilation resulting from chronic necrotizing infection is called bronchiectasis.

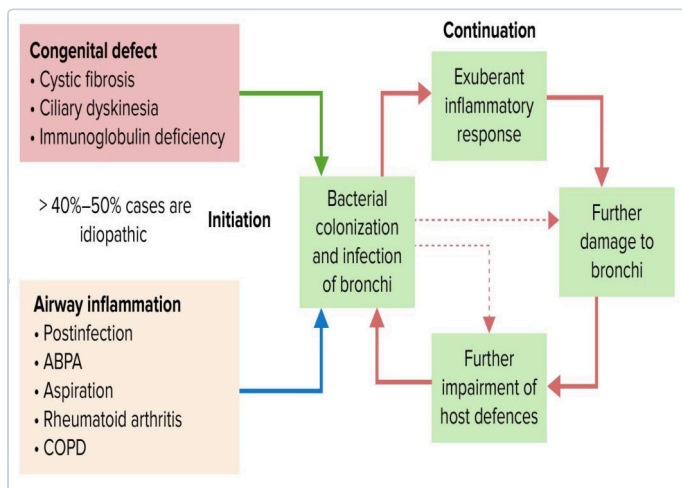
Cause



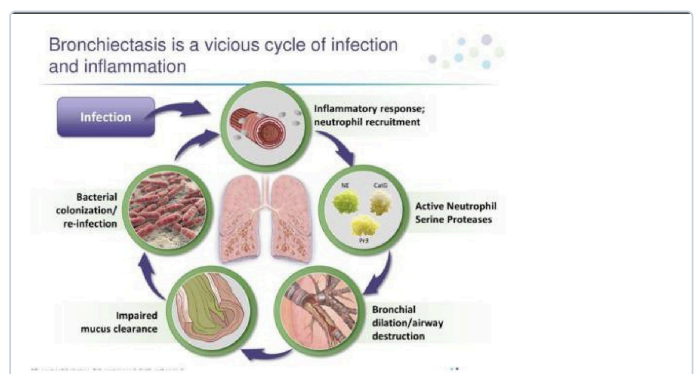
Slide 16: Causes: obstruction, congenital, necrotising pneumonia, miscellaneous

2.9 Pathophysiology: The Vicious Cycle (Slides 17-18)

★★★★ **Initiation: congenital defect** (CF, ciliary dyskinesia, Ig deficiency) or **airway inflammation** (post-infection, ABPA, aspiration, RA, COPD) → **bacterial colonisation and infection of bronchi** → ★ **Continuation (vicious cycle): exuberant inflammatory response** (neutrophils, serine proteases) → **further bronchial damage** → **further impairment of host defences** (poor mucus clearance) → **more colonisation** → round again. *[added: this is Cole's "vicious cycle" hypothesis.]*



Slide 17: Initiation and continuation of the vicious cycle

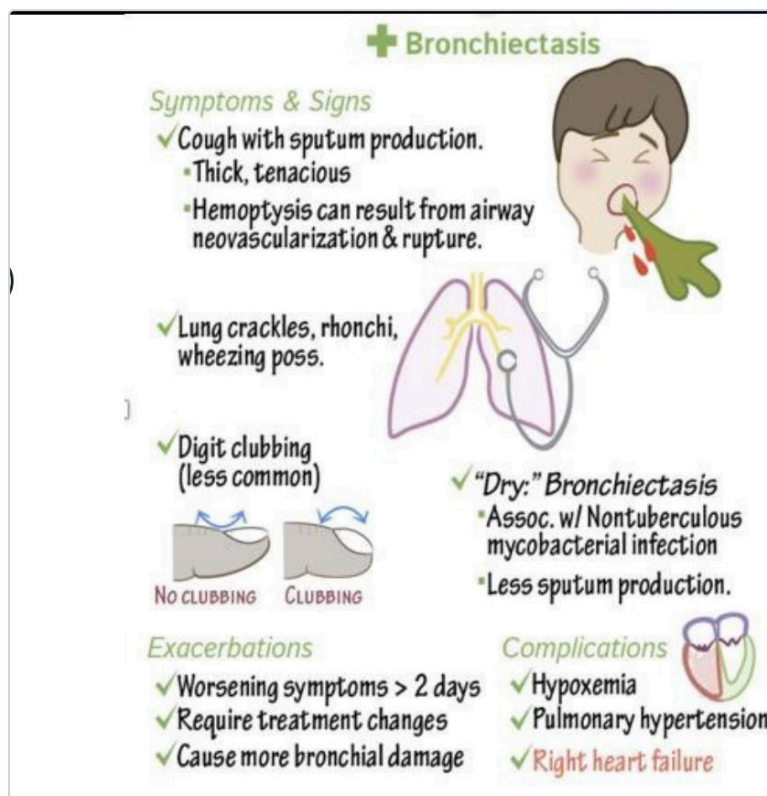


Slide 18: Infection ↔ inflammation cycle

2.10 Clinical Features (Slides 19-20)

- ★ **Chronic cough with LARGE amounts of mucopurulent sputum**
- Dyspnoea**
- ★ **Haemoptysis: mild, self-limiting** (can be major from bronchial vessels *[added]*)
- Non-specific: **fatigue, weight loss, pallor** (anaemia)
- Clubbing: uncommon** (per slide)

- ★ **Exacerbations:** recurrent pneumonia/acute infection of dilated bronchi, ★ **often** ***Pseudomonas aeruginosa***; ↑ sputum above baseline, **low-grade fever** (figure: worsening > 2 days, needing treatment change)
- Auscultation:** ★ **coarse crackles and rhonchi**; **wheeze** (secretions, airway collapse, or coexisting disease)
- Complications (figure): **hypoxaemia, pulmonary hypertension, right heart failure**
- "Dry" bronchiectasis (figure): **non-tuberculous mycobacteria**, little sputum

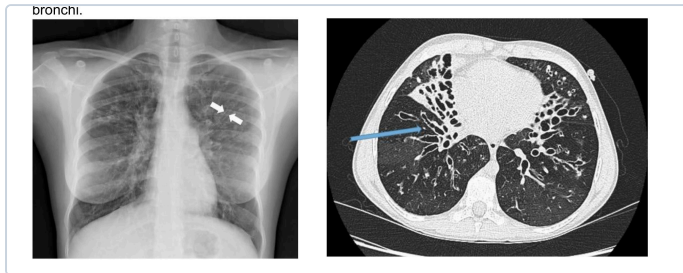


Slide 19: Bronchiectasis: symptoms, exacerbations, complications

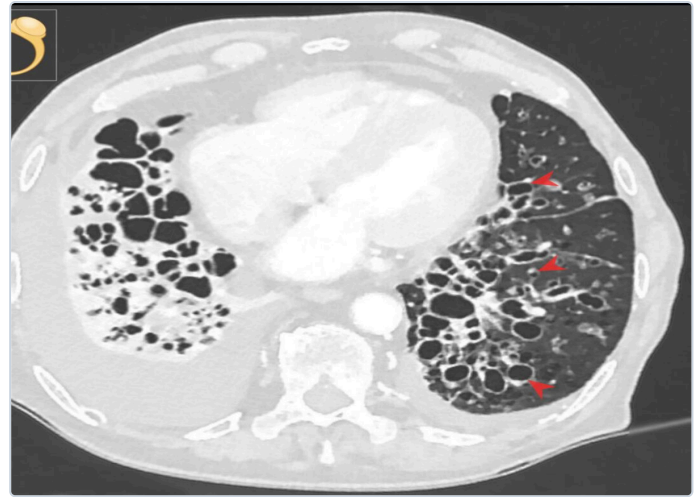
2.11 Diagnostics (Slides 21-24)

★★★★ **High-resolution CT (HRCT) = the diagnostic test:**

Sign	★ Meaning
★ Tram-track sign	Parallel lines of thickened bronchial walls around lucent air in dilated bronchi (bronchi that don't taper)
★ Signet-ring sign	Cross-section of a dilated bronchus larger than its adjacent pulmonary artery (the artery is the "jewel")



Slide 22: Tram-track sign on CXR and CT



Slide 23: Signet-ring sign on CT: bronchi larger than adjacent arteries

- **CXR:** abnormal but **non-specific**; may miss mild cases
- **PFTs:** obstructive pattern (not needed for diagnosis; useful to **monitor**; may be obstructive, restrictive or mixed)
- **Sputum culture** (induced sputum or BAL)

★★ Find the cause (Slide 24): often **idiopathic**

- **CBC:** ↓ Hb, variable WBC, ↑ platelets
- **ABG:** may show **hypercapnia**
- ★ **Serum immunoglobulins + electrophoresis** → exclude **immunodeficiency**
- ★ **Aspergillus fumigatus IgG and IgE** → exclude **ABPA**
- ★ **Serum AAT** → **α1-antitrypsin deficiency**
- [added: sweat chloride / CFTR genetics if CF suspected (younger patients)]

2.12 Management (Slide 25)

- ★ **Antibiotics:** suppress bacteria, reduce symptoms and exacerbations. ★ **Secondary prevention if ≥ 3 exacerbations/year:** give **for ≥ 3 months**, extend by response
- ★ **Vaccination:** influenza + pneumococcal
- ★ **Chest physiotherapy** (airway clearance)
- **Mucolytics**
- **Bronchodilators:** **limited benefit**, case-by-case
- **Oxygen therapy**
- ★ **Surgical resection: lobectomy for bronchial haemorrhage** (localised disease)

BRONCHIECTASIS

Bronchiectasis is a destructive lung disease characterized by chronic (permanent/irreversible) dilatation of the bronchi.

Symptoms:

General

- Weight loss
- Poor Appetite
- Weakness
- Clubbing nails
- Drum Stick Nail

Lungs

- Paroxysmal Cough
- Cough purulent
- Dyspnoea
- Wheezing, Crepitation from lungs
- Blood in Cough
- Cough aggravated in morning
- Offensive cough by change of position

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CAUSES	DIET & REGIMEN	TREATMENT	MEDICINES
Cystic fibrosis Airway obstruction Lung Infections & injury Impaired host defences Autoimmune diseases Congenital Cigarette smoking	• Avoidance of smoking • Adequate hydration • Adequate antioxidant • Small, frequent feedings	• Postural drainage • Antibiotic therapy • Treatment for co-infection	• Aconite • Belladonna • Bryonia alba • Hepar Sulphur • Kali Bichromicum • Phosphorus

Slide 38: Bronchiectasis summary card

🔄 **Recap: Bronchiectasis.** Permanently dilated, destroyed bronchi from infection/inflammation + poor clearance (vicious cycle); 40-50% idiopathic; CF, ciliary dyskinesia, Kartagener, ABPA, immunodeficiency, COPD, AAT deficiency. Large volumes of daily purulent sputum, crackles, haemoptysis, Pseudomonas exacerbations. HRCT: tram-track + signet ring. Look for the cause (Igs, Aspergillus serology, AAT). Physio, vaccines, mucolytics; long-term antibiotics if ≥ 3 exacerbations/year; lobectomy for bleeding.

🔗 **Link:** The most important genetic cause of bronchiectasis is CF, which affects every exocrine gland, not only the lungs.

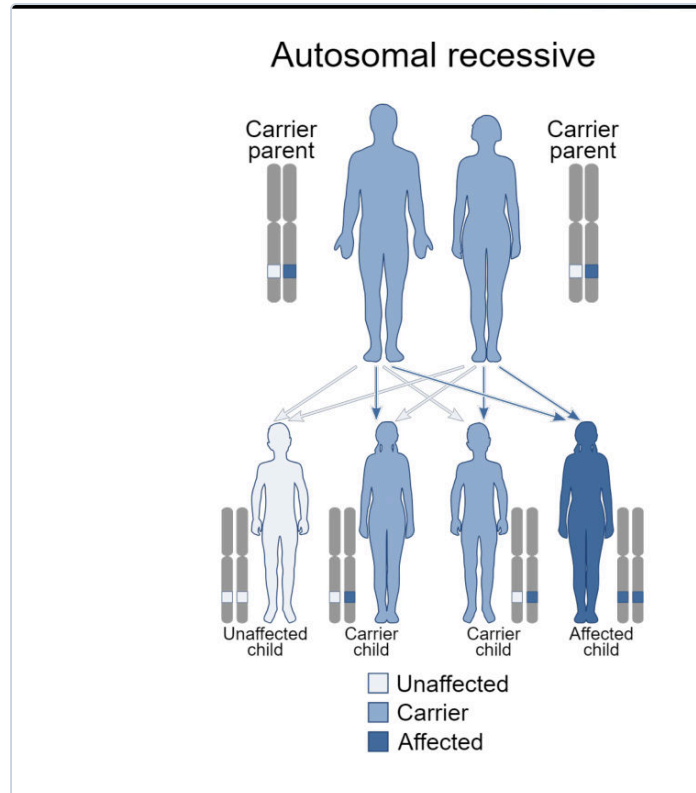
👉 **Up next: Cystic fibrosis.**

2.13 Cystic Fibrosis: Definition & Genetics (Slides 26-27)

★ **Definition:** a defect in a **chloride-channel protein** → impaired **Cl⁻ and water transport** → ★ **thick, viscous secretions** in the **respiratory tract, exocrine pancreas, sweat glands, intestines and genitourinary tract.**

★★ **Genetics:**

- ★ **Autosomal recessive**
- ★ **CFTR** (cystic fibrosis transmembrane conductance regulator) gene mutations
- ★ Most common: **codon deletion → absent phenylalanine at position 508 (ΔF508)**



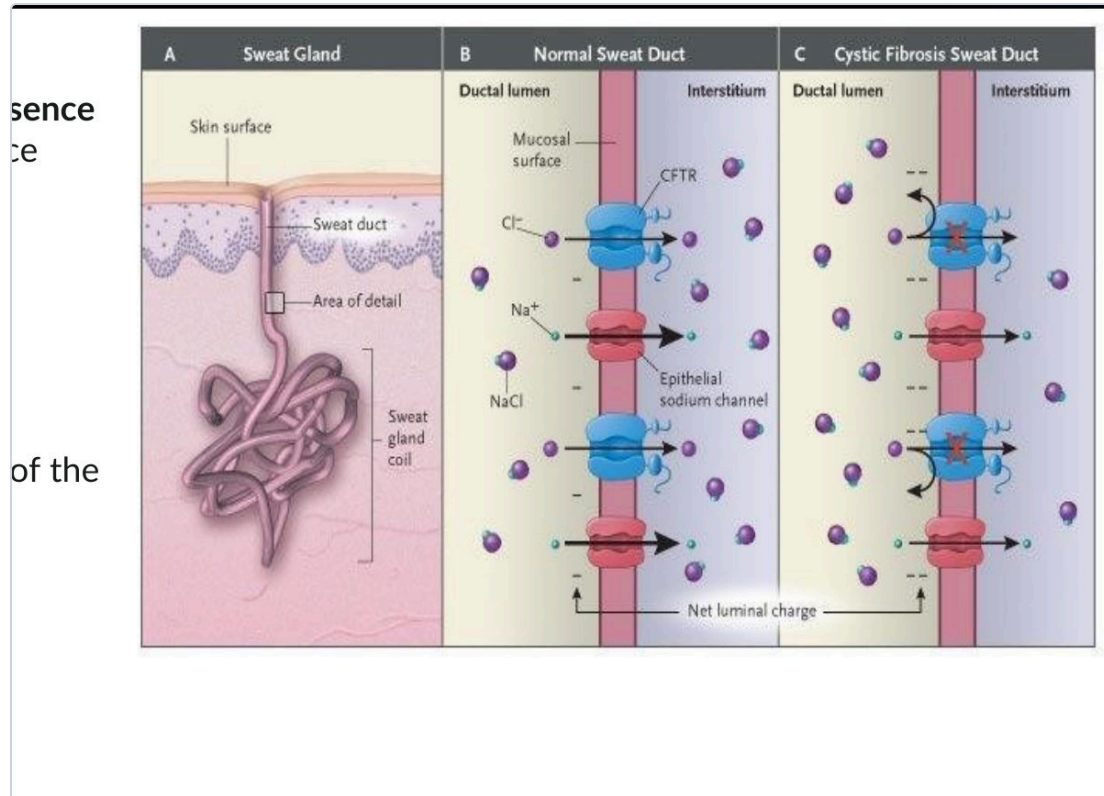
Slide 27: Autosomal recessive inheritance (two carrier parents)

2.14 Pathophysiology: Sweat vs Other Glands (Slides 28-29)

★ Mutated CFTR → **misfolded protein** → **no ATP-gated Cl^- channel** on epithelial surfaces.

★★★★ **The channel runs in opposite directions in different glands:**

	Sweat glands	All other exocrine glands (lungs, GI)
Normal CFTR job	Cl^- lumen → cell (reabsorption)	Cl^- cell → lumen (secretion)
In CF	Can't reabsorb Cl^- → less Na^+ and water reabsorbed	Can't secrete Cl^- (and water) → intracellular Cl^- builds up → ↑ Na^+ and water reabsorption
Result	★ Salty sweat (↑ NaCl) → basis of the sweat test	★ Dehydrated, hyperviscous mucus → blocks small passages

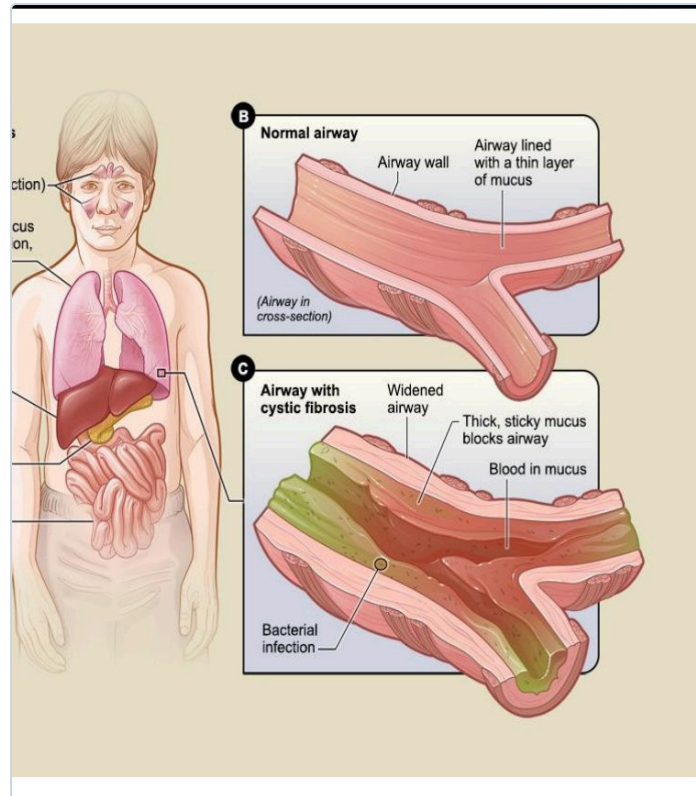


Slide 28: Sweat duct: normal vs CF

2.15 Clinical Features by System (Slides 30-33)

★ Respiratory (common in adulthood):

- **Chronic obstructive lung disease with bronchiectasis**
- **Chronic sinusitis, ★ nasal polyps**
- Recurrent productive cough and infections; ★ susceptibility to **Pseudomonas aeruginosa, Aspergillus**
- Wheeze/dyspnoea (obstruction, hyperreactivity)
- **Barrel chest**, moist rales, **hyperresonance**
- **Haemoptysis**
- ★ **Clubbing** (chronic hypoxia)

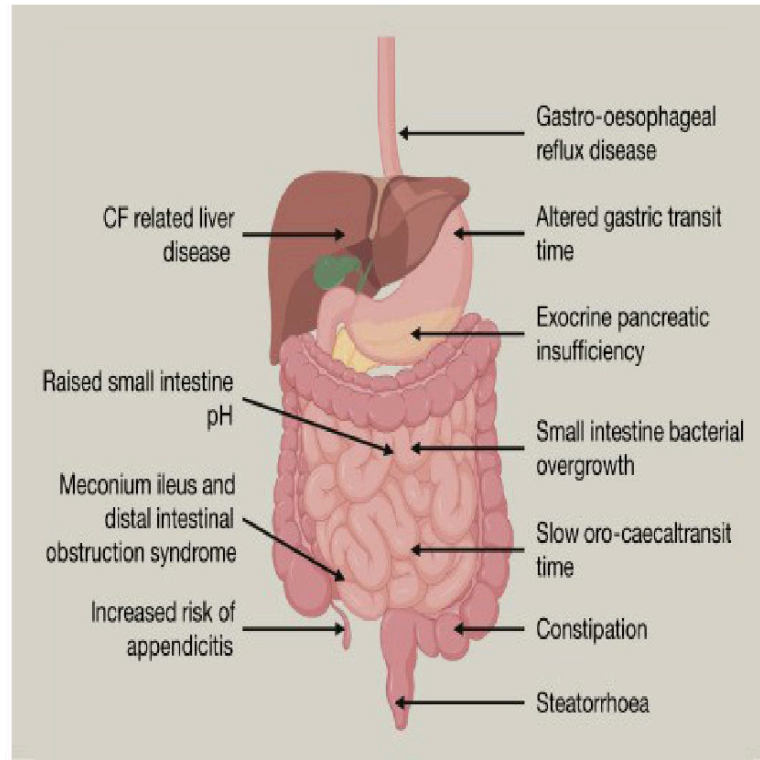


Slide 30: CF airway: thick sticky mucus, inflammation, bacterial infection

★ **Gastrointestinal** (common in **children**):

- ★ **Meconium ileus** (newborn)
- ★ **Failure to thrive**
- **Pancreas:** pancreatitis; ★ **CF-related diabetes**; ★ **exocrine pancreatic insufficiency** → **foul-smelling steatorrhoea**, ★ **fat-soluble vitamin (A, D, E, K) deficiency**, malabsorption, abdominal distension, diarrhoea, **hypoproteinaemia**
- **Liver/biliary:** gallstones, **cholestasis**, **fatty liver** → **cirrhosis**, ★ **biliary cirrhosis with portal hypertension**
- **Intestinal obstruction** (distension, pain, palpable mass) *[added: distal intestinal obstruction syndrome]*
- ★ **Rectal prolapse**

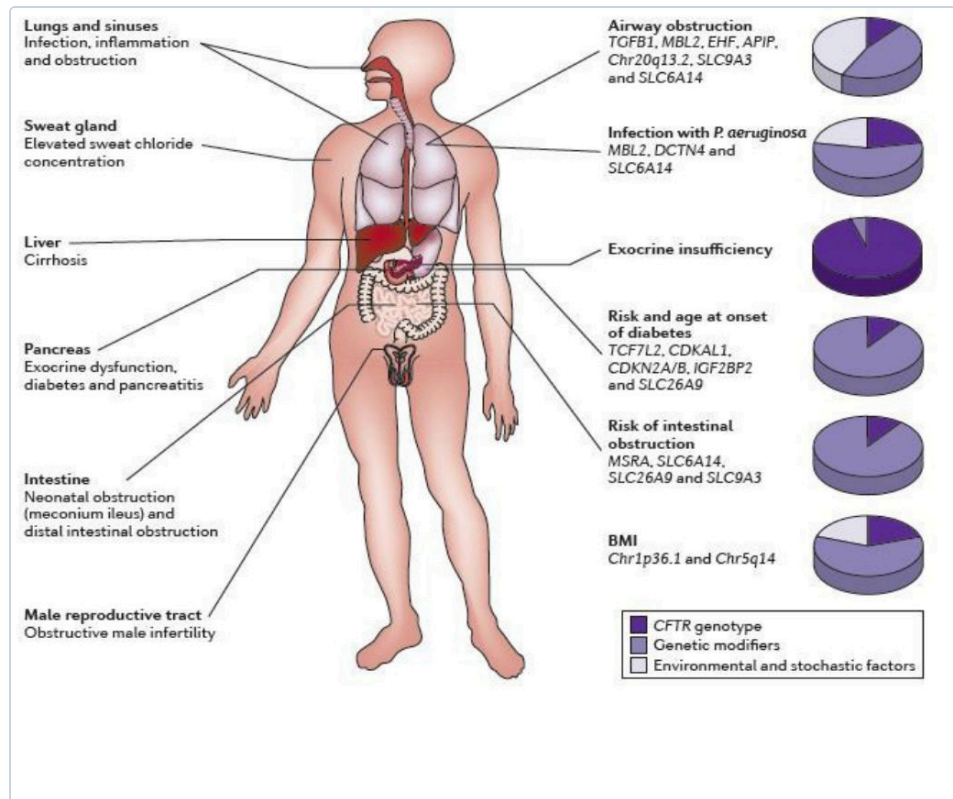
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Slide 31: CF gastrointestinal involvement

Other systems:

System	Features
Sweat	★ Very salty sweat ; possible electrolyte wasting
Musculoskeletal	Fractures from osteopenia ; kyphoscoliosis
Urinary	Nephrolithiasis, nephrocalcinosis ; frequent UTIs
★ Male genital	★ Usually infertile: obstructive azoospermia (spermatogenesis may be intact); ★ absent vas deferens ; undescended testis
Female genital	Reduced fertility (viscous cervical mucus); menstrual abnormalities (amenorrhoea)
Development	Delayed secondary sexual characteristics



Slide 33: CF organ involvement

2.16 Diagnostics (Slide 34)

- ★★★ **Sweat chloride test:** ≥ 60 mmol/L (diagnostic)
- ★ **CFTR genotyping**
- Nasal potential difference** (Cl^- movement)
- CXR** (central cystic bronchiectasis, mucus plugging)
- PFTs:** obstructive pattern

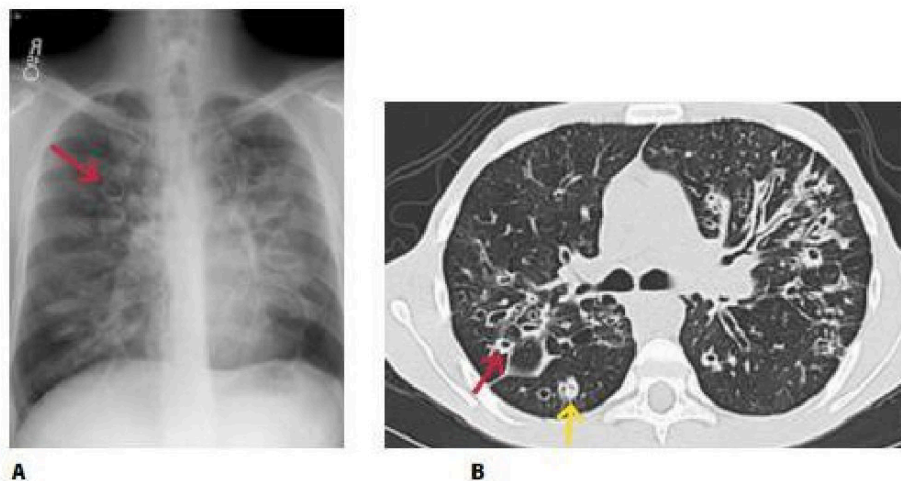


FIGURE 4.5. Cystic fibrosis. (A) Frontal CXR showing central cystic bronchiectasis (arrow) in a patient with CF. (B) Transaxial CT image showing cystic bronchiectasis (red arrow), with some bronchi containing impacted mucus (yellow arrow). (Reproduced with permission from USMLE-Rx.com.)

Slide 34: CF imaging: central cystic bronchiectasis and impacted mucus

2.17 Management (Slides 35-36)

★ Pharmacological:

Therapy	Detail
High-dose ibuprofen	Slows lung decline; consider in 6-17 y with mild disease
Bronchodilators (SABA, LABA)	Before inhaled therapy / physio, or for hyperresponsiveness; not routine long-term
★ Mucolytics	★ Hypertonic saline (6-7%) nebulised (osmotic + mucociliary effect) · ★ Dornase alfa (aerosolised recombinant DNase : breaks down DNA in mucus)
★ Antibiotics	For infections, ★ especially Pseudomonas
Oxygen	Intermittent : desaturation with exercise/sleep · Long-term : hypoxia on room air, or borderline hypoxia + cor pulmonale
★ Double-lung transplant	End-stage
<i>CFTR modulators</i>	<i>Named in the objectives only [added: e.g., ivacaftor; elexacaftor-tezacaftor-ivacaftor for $\Delta F508$]</i>

★ Non-pharmacological:

- ★ **Airway clearance techniques: mainstay** of CF care
- Vaccinations** (influenza, pneumococcal)
- ★ **Chest physiotherapy**: postural drainage with percussion/clapping
- Exercise** (swimming, jogging, cycling)

★ **Treat other systems: ★ pancreatic enzyme replacement; ★ fat-soluble vitamin supplements**

Cystic fibrosis

Epidemiology
 1:3000–1:6000 in populations of northern descent
 Reduced life expectancy (~45 years)

Genetics
 Autosomal recessive disorder
 (most commonly a $\Delta F508$ Phenylalanine deletion) in the CFTR gene on the long arm of chromosome 7 that causes chloride channel defects in exocrine glands

Diagnosis
 Suspicion for CF (positive newborn screen, family history, or symptoms) AND laboratory evidence of CF (sweat chloride testing, genetic analysis, or both)

Management
 CFTR modulator therapy
 Airway clearance and pharmacological preservation
 Antibiotics for chronic lung infections and acute exacerbations
 Support of nutrition (e.g., pancreatic enzyme supplements)
 Lung transplant

Complications
 Progressive respiratory decline and, ultimately, respiratory failure
 Chronic lung infections (e.g., *Pseudomonas aeruginosa*)
 Systemic diseases: liver disease (CFLD) and diabetes (CFRD)
 Gastrointestinal obstruction (e.g., SBO, DIOS)

Neonatal symptoms

Meconium ileus
 Malabsorption/failure to thrive
 Neonatal obstructive jaundice

Slide 39: CF summary card

 **Recap: CF.** Autosomal recessive CFTR defect ($\Delta F508$). Sweat glands can't reabsorb Cl^- (salty sweat); other glands can't secrete it (thick mucus). Lungs: bronchiectasis, Pseudomonas, polyps, clubbing. GI: meconium ileus, failure to thrive, steatorrhea, ADEK deficiency, CF diabetes, biliary cirrhosis, rectal prolapse. Infertile men (absent vas). Sweat $\text{Cl}^- \geq 60$. Airway clearance, DNase, hypertonic saline, anti-Pseudomonas antibiotics, enzymes and vitamins, transplant.

3. High-Yield Comparison Tables

Table A: The three diseases (see the added diagram at the top)

Table B: CFTR in sweat vs airway (see §2.14)

Table C: Imaging signs

Sign	Disease
Thick-walled cavity + air-fluid level	Lung abscess
Tram-track (parallel thick walls)	Bronchiectasis
Signet ring (bronchus > artery)	Bronchiectasis
Central cystic bronchiectasis + mucus plugs	CF

Table D: Bronchiectasis work-up

Test	Looks for
Immunoglobulins + electrophoresis	Humoral immunodeficiency
Aspergillus IgG/IgE	ABPA
Serum AAT	$\alpha 1$ -antitrypsin deficiency
Sweat chloride / genetics [added]	CF
Sputum culture	Pseudomonas, NTM

Table E: Antibiotic rules

Disease	Antibiotic rule
Lung abscess	Anaerobic cover (amp-sulbactam, carbapenem, clindamycin), 3-6 weeks ; add S. aureus cover in IVDU
Bronchiectasis	Treat exacerbations (often Pseudomonas); ≥ 3 months prophylaxis if ≥ 3 exacerbations/yr
CF	Anti-Pseudomonas

4. Rules, Exceptions, Patterns

- **Abscess = oral anaerobes via aspiration** until proven otherwise; **multiple bilateral = haematogenous**.
- **Blood cultures are often negative in anaerobic abscess**; culture sputum/BAL.
- **Start antibiotics immediately after sampling**; don't wait.
- **CT with contrast confirms an abscess; HRCT confirms bronchiectasis**.
- **Drainage, not surgery, is first-line intervention** for an abscess.

- **Bronchiectasis needs infection/inflammation PLUS impaired clearance/defence.**
- **40-50% of bronchiectasis is idiopathic.**
- **Pseudomonas = the exacerbation organism** in bronchiectasis and CF.
- **CFTR reabsorbs Cl^- in sweat ducts but secretes Cl^- elsewhere**, hence salty sweat but dry mucus.
- **Sweat $\text{Cl}^- \geq 60 \text{ mmol/L} = \text{CF}$.**
- **Lungs dominate in adults; GI dominates in children** with CF.
- **Male CF infertility is obstructive** (absent vas); spermatogenesis may be normal.

5. Clinical Correlations

- **Alcoholic or post-stroke patient with poor teeth and foul sputum** → aspiration lung abscess.
- **IV drug user with multiple bilateral cavities** → septic emboli (right-sided endocarditis) → echocardiogram, *S. aureus* cover.
- **Abscess not resolving on antibiotics** → bronchoscopy for **cancer or foreign body**.
- **Situs inversus + chronic sinusitis + bronchiectasis + male infertility** → Kartagener syndrome.
- **Asthmatic with worsening wheeze, eosinophilia, central bronchiectasis** → ABPA (*Aspergillus* IgE) [central detail added].
- **Newborn with meconium ileus** → CF until proven otherwise.
- **Child with recurrent pneumonia + greasy stools + poor growth** → sweat test (Case 2).
- **Young man with azoospermia and absent vas deferens** → check CFTR.
- **Massive haemoptysis in bronchiectasis** → bronchial artery embolisation [added] or lobectomy (slide).

6. Rapid Revision (Last-Day)

- Suppurative lung disease: **abscess, bronchiectasis, CF, pneumonia**.
- **Abscess** = localised pus + necrosis. Primary (aspiration, normal lung) vs secondary (immunocompromise, cancer, septic emboli/IVDU).
- Organisms: **oral anaerobes** (*Peptostreptococcus*, *Prevotella*, *Bacteroides*, *Fusobacterium*); aerobes (*S. aureus*, *Klebsiella*, *Pseudomonas*, *Strep*); **E. histolytica**; fungi.
- Risks: impaired consciousness/swallowing, **poor dentition**, obstruction (cancer, foreign body), immunocompromise, CF.
- Sequence: aspiration → pneumonitis → vascular obstruction → necrosis/liquefaction → abscess (7–14 days).
- Acute < 4–6 weeks; chronic > 6 weeks. **Fever, foul purulent sputum** (↑ lying on healthy side), night sweats, haemoptysis, weight loss, pleuritic pain; **clubbing if chronic**.
- ↑ WBC, CRP, PCT; **blood cultures often negative**; sputum/BAL culture; **start empirical antibiotics immediately**; **CT with contrast confirms**; **cavity + air-fluid level**.
- Location: aspiration → right; prone → RML (slide); supine → posterior upper / superior lower segments; haematogenous → **bilateral, multiple**.
- Rx: **ampicillin-sulbactam / carbapenem / clindamycin, 3-6 weeks**; drainage if large, bleeding or failing; surgery rarely.
- **Bronchiectasis** = permanent dilatation/destruction; infection/inflammation + poor clearance; **vicious cycle**; **40-50% idiopathic**.
- Causes: recurrent infection, **CF, PCD, ABPA, Kartagener**, smoking, **COPD, aspiration, tumour, AAT deficiency**, immunodeficiency, RA/Sjögren/Crohn.

- **Large daily mucopurulent sputum**, dyspnoea, mild haemoptysis, crackles/rhonchi; clubbing uncommon; exacerbations (**Pseudomonas**).
- **HRCT: tram-track + signet ring**. Work-up: **Igs, Aspergillus IgG/IgE, AAT**.
- Rx: **antibiotics ≥ 3 mo if ≥ 3 exacerbations/yr**, vaccines, **physio**, mucolytics, O₂, **lobectomy for bleeding**.
- **CF** = AR, **CFTR $\Delta F508$** , misfolded ATP-gated Cl⁻ channel. Sweat: can't reabsorb → **salty**. Airway/GI: can't secrete → **thick mucus**.
- CF lungs: bronchiectasis, **Pseudomonas/Aspergillus**, sinusitis, **nasal polyps**, clubbing. GI: **meconium ileus**, failure to thrive, **steatorrhoea, ADEK deficiency**, CF diabetes, biliary cirrhosis, **rectal prolapse**. Male **infertility (absent vas)**.
- **Sweat Cl⁻ ≥ 60 mmol/L**; genotyping; nasal potential difference.
- CF Rx: **airway clearance (mainstay)**, **hypertonic saline, dornase alfa**, anti-Pseudomonas antibiotics, ibuprofen (6–17 y), O₂, **enzymes + ADEK**, vaccines, **double-lung transplant**.

7. Exam-Style Questions

Q1 (MCQ, lecture Case 1, Slide 40). 65 ♂, long-standing productive cough, worse since a **stroke 3 years ago**; 40 pack-years; now **copious blood-tinged sputum**, fever 38 °C, SpO₂ 92%, ★ **clubbing, coarse crackles**. CXR: ★ **tram-track opacities** in the right lower field. Diagnosis? **a. Bronchiectasis** b. Chronic bronchitis c. Aspiration pneumonia d. Emphysema e. PE f. Lung cancer
Answer: a. [reasoning added; no key on the slide] **Tram-track sign** + chronic copious sputum + haemoptysis + crackles = bronchiectasis. Recurrent **aspiration after the stroke** (right lower lobe) is the likely cause. Chronic bronchitis and emphysema don't give tram tracks; aspiration pneumonia doesn't explain years of symptoms.

Q2 (MCQ, lecture Case 2, Slide 41). 2-year-old: productive cough, **4 pneumonias in a year, foul-smelling, bulky, greasy stools, 3rd percentile height, 5th weight**. Best initial test for the cause? **e. Sweat chloride test** **Answer: e.** Recurrent chest infections + steatorrhoea (pancreatic insufficiency) + failure to thrive = **cystic fibrosis** → **sweat chloride (≥ 60 mmol/L)** first. Genetic testing confirms or follows [ordering added]. Immunoglobulins would suit immunodeficiency, but that doesn't explain steatorrhoea.

Q3 (MCQ). An alcoholic man has fever, foul-smelling sputum and a thick-walled cavity with an air-fluid level. Best initial antibiotic? a) Azithromycin **b) Ampicillin-sulbactam** c) Ciprofloxacin d) Vancomycin alone **Answer: b** (anaerobic cover; clindamycin if penicillin-allergic).

Q4 (MCQ). On HRCT, a bronchus appears larger than its accompanying pulmonary artery. This is the: a) Tram-track sign **b) Signet-ring sign** c) Air-crescent sign d) Halo sign **Answer: b.**

Q5 (Short answer). Explain why CF patients have salty sweat but thick airway mucus. **Answer:** CFTR normally **reabsorbs** Cl⁻ from the **sweat duct lumen** but **secretes** Cl⁻ into the lumen in **airway/GI glands**. Without it, sweat Cl⁻ (and Na⁺) **can't be reabsorbed** → salty sweat. In the airways Cl⁻ and water **aren't secreted** while Na⁺ and water reabsorption rises → **dehydrated, viscous mucus**.

8. Exam Pitfalls & Traps

1. **Blood cultures are usually positive in lung abscess.** Often **negative** with anaerobes; use sputum/BAL.
2. **Waiting for culture before antibiotics.** Start **empirically** after sampling.
3. **Surgery first for an abscess.** First-line intervention is **drainage**; surgery is rare.

4. **Haematogenous abscesses are single.** They're **bilateral and multiple**.
5. **CXR diagnoses bronchiectasis.** It's non-specific; **HRCT** is needed.
6. **Clubbing is common in bronchiectasis.** The slide says **uncommon**; it's typical of chronic abscess and CF.
7. **CFTR direction:** reabsorbs Cl^- in **sweat** but secretes it in **airways/GI**.
8. **CF male infertility = no sperm production.** It's **obstructive** (absent vas); spermatogenesis may be intact.
9. **Bronchodilators are core in bronchiectasis/CF. Limited benefit;** not routine long-term.
10. **⚠ Location slide:** the lecture says aspiration while **prone → right middle lobe**. Many texts teach **upright → basal segments of the right lower lobe** and **supine → posterior upper / superior lower lobe segments**. Answer per the lecture, but know the standard teaching.
11. **⚠ Antibiotic thresholds:** bronchiectasis prophylaxis is for **≥ 3 exacerbations/year**, for **≥ 3 months**. Don't confuse with abscess treatment (3–6 weeks).
12. **⚠ Slide typo:** "Bronchiactasis" in the file title = bronchiectasis. **CFTR modulators** appear in the objectives but not on any slide.